Urolithiasis (2022) 50:599–610
https://doi.org/10.1007/s00240-022-01341-2

ORIGINAL ARTICLE

Predicting narrow ureters before ureteroscopic lithotripsy with a neural network: a retrospective bicenter study

Jun Wang[1] · Dawei Wang[2] · Yong Wang[3] · Shoutong Wang[1] · Yi Shao[1] · Jun Lu[1]

Received: 8 April 2022 / Accepted: 30 May 2022 / Published online: 23 June 2022

Abstract
In some patients, the passage of semi-rigid ureteroscopes up the ureter is impossible due to narrow ureteral lumen. We established a neural network to predict the inability of the ureter to accommodate the semi-rigid ureteroscope and the need for active or passive dilatation using non-contrast computed tomography (CT) images. Data were collected retrospectively from two centers of 1989 eligible patients who underwent ureteroscopic lithotripsy with ureteral stones. Patients were categorized into two groups: control and narrow ureter. The network was designed and trained for predicting a narrow ureter during initial ureteroscopic lithotripsy, which integrated multi-scale features of the ureter. The predictive efficacy of neural networks DenseNet3D, ResNet3D, ResNet3D MC, and TimeSformer was compared. Furthermore, a previous ureteroscopy or a history of double-J stent placement, ureteral wall thickness and Hounsfield unit (HU) density of the ureter under the stone were compared. Model performance was assessed based on the accuracy, area under the receiver operating characteristic curve (AUC ROC), etc. The DenseNet3D-based network achieved an AUC ROC score of 0.884 and an accuracy of 85.29%, followed by the ResNet3D-based network, the ResNet3D MC-based network, and the TimeSformer-based network. The DenseNet3D-based network significantly outperformed other candidate predictors. Furthermore, the networks were validated in an external test set. Decision curve analysis showed the clinical utility of the neural network. The neural network provides an individualized preoperative prediction of narrow ureter based on non-contrast CT images, which could be employed as part of a surgical decision-making support system.

Keywords Neural network · Ureteroscopic lithotripsy · Ureteral stone · Artificial intelligence · Difficult ureter

Jun Wang, Dawei Wang, Yong Wang these authors contributed equally to the article.

✉ Yi Shao
drshaoyi@163.com

✉ Jun Lu
842320@163.com

Jun Wang
linkwj@126.com

Dawei Wang
wdwrjhn@163.com

Yong Wang
wyarmy@163.com

Shoutong Wang
shoutongwang@163.com

[1] Department of Urology, Shanghai General Hospital, School of Medicine, Shanghai Jiao Tong University, No. 100 Hai Nin Road, Shanghai 200080, China

[2] Department of Urology, Ruijin Hospital, School of Medicine, Shanghai Jiao Tong University, Shanghai 200025, China

[3] Department of Urology, Jiangqiao Hospital in Jiading District of Shanghai, Shanghai 200000, China

Abbreviations

AUC	Area under the curve
CT	Computed tomography
CTU	Computerized tomography urogram
HU	Hounsfield unit
ROC	Receiver operating characteristic curve
SD	Standard deviation

Introduction

Ureteral stone is one of the most common urological conditions. Although ureteroscopy is a generally safe and effective procedure for removing ureter stones, with a 90% clearance rate, some ureters can be difficult to access due to narrow ureteral lumens [1, 2]. Placing a ureteral stent for passive dilatation or performing active ureteral dilatation is required [3].

When presenting the different treatment options to a patient with a ureteral stone, the chances of a failed ureteroscopy, the need for a two-stage procedure, or the potential risk and cost of various balloon or coaxial dilatation instruments are an integral part of the discussion [4]. However, nowadays we still have no reliable radiographic method to evaluate the narrow ureter preoperatively. Checking computed tomography (CT) images with the naked eye cannot reliably distinguish those ureters. Some reported variables are only available in computerized tomography urogram (CTU) that are not part of a preoperative protocol for distal ureteral stone surgery [3].

With the advancement of deep-learning technology in recent years, neural network tools have been developed in urolithiasis to detect ureteral stones and predict spontaneous passage [5–7]. A neural network mimics the human brain. Like teaching to recognize objects, a neural network implicitly learns to classify objects through appropriate algorithms after training with numerous images and corresponding labels. To the best of our knowledge, no previous research evaluates a neural network method for predicting narrow ureter.

The use of neural networks was explored using non-contrast CT images to predict narrow ureter for a semi-rigid ureteroscope. As previous studies have shown that clinical parameters include a previous ureteroscopy or a history of double-J stent placement, and older age is associated with a higher success rate of upper urinary tract access, we selected them as predictors for comparison [2, 4]. Ureteral wall thickness and Hounsfield unit (HU) density of the ureter under the stone were also selected as candidate predictors, because they were associated with the success of extracorporeal shock wave lithotripsy treatment or stone impaction [8–14]. The efficacy of different state-of-the-art deep-learning algorithms was assessed. Moreover, we validated our model in an external dataset.

Patients and methods

The dataset

This retrospective investigation was authorized by each institutional research ethics board, which waived informed consent. There was no usage of protected health information. We examined 2168 consecutive surgical patients from Shanghai General Hospital who had ureteral stones between April 2013 and April 2021, and 81 patients (1 patient had bilateral ureteral stones) from Jiangqiao Hospital in Jiading district of Shanghai between Oct 2020 and March 2022 (Fig. 1). Patients who met the following criteria were included in the analysis: 1. ureteral stones; 2. planned ureteral lithotripsy; 3. surgery performed by senior urology endoscopists (with least 3 years experience); 4. complete records and CT scan data (within 1 month before surgery). Besides, patients who had the following conditions were excluded from the analysis: fever patients (within 1 month before surgery), positive urine culture (untreated or not turned negative after antibiotic treatment), already had stent placement, history of ureter stricture, pregnancy, and known ureteral anatomic abnormalities (Fig. 1).

A total of 1907 and 82 scans with ureter stone were included in the newly created data bank from Shanghai General Hospital and Jiangqiao Hospital in Jiading district of Shanghai (Fig. 1). If one patient performed lithotripsy on one side and stent placement on the other side, only the data from the lithotripsy side were included. If there were multiple ureteric stones in one ureter, only the lowest stone was extracted for analysis. The narrow ureter was defined as ureters that could not accommodate a ureteroscope due to narrow ureteral lumens and stones that cannot be accessed during the initial ureteroscope examination without resorting to other forms of active or passive dilatation. Images were categorized into control and narrow ureter groups according to the occurrence of the narrow ureter (Fig. 1). Surgical details, instruments, CT scanners and CT parameters are listed in Supplemental Table 1. All CT images were resampled to a uniform slice thickness using trilinear interpolation.

The dataset from the Shanghai General Hospital cohort was randomly divided into three sets (training, validation, and test) without overlapping. The datasets from Jiangqiao Hospital were used as an external test set. The training dataset was used to build the neural network model, the validation dataset was employed to evaluate the developed model, and the performance of the model was evaluated using the independent test datasets (Fig. 1). In detail, as the control group had more cases than the narrow ureter group, the narrow ureter group was randomly divided into these three sets according to the ratio of 8:1:1. The control group was also divided into three groups at random, but the number of validation and test cases was the same as the narrow ureter group. The training set in the narrow ureter group was oversampled with rotation, horizontal, or vertical shift to generate a new training set with the same size as the control training set. Training sets from both groups were augmented with random flip, random affine, random noise, and random elastic deformation in the training set. Low frequent data (proximal ureteral stone, ureteral flexion) were also oversampled with data augmentation. No additional augmentation was performed on the validation dataset and test dataset.

Neural network

It was assumed that both local detailed ureteral features and global contextual information such as hydronephrosis and stone position are required for accurate classification. The network was designed with three inputs. Vesicoureteral junction and peri-stone ureter inputs provided the local ureteral

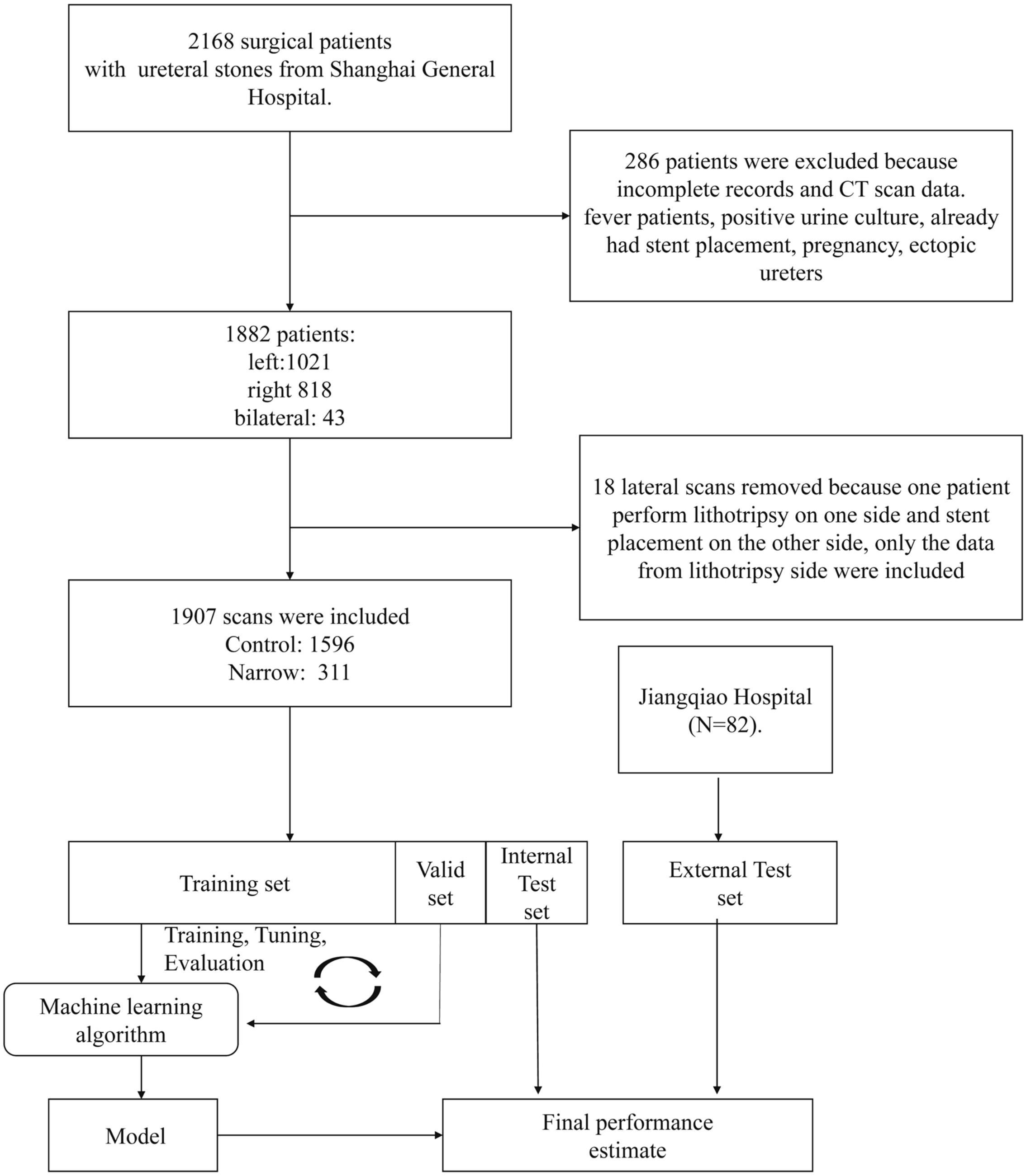

Fig. 1 Flowchart of this study

texture features (Fig. 2), based on the fact that the ureteroscope is more likely to be stuck at intramural ureters or beneath stones. The A cube input containing the entire ureter and surrounding tissue was created to provide global contextual information. We called it the entire ureter input (Fig. 2). Detailed information of three inputs (proposed coordinates, included structures, cube size, and resize) are listed in Supplemental Table 2. In brief, given that the ureter arises from the renal pelvis, passes over (anterior to) the psoas muscle, and enters the bladder on the posterior bladder aspect in the trigone, the coordinates of the vesicoureteral junction and the entire ureter were determined using the bladder and psoas muscle anatomical information. The morphological information of the bladder and psoas muscle was obtained from ensembled abdominal segmentation using OBELISK-Net [15] and U-Net [16] (Fig. 2). The ureteral stone was segmented using the method described by Jendeberg et al. [5]. All three inputs were initially automatically labeled using the aforementioned proposed coordinates and manually checked by one radiologist.

As stones are frequently stuck in the intramural ureter, one of the narrowest regions of the ureter, the vesicoureteral junction input and the peri-stone ureter input can partially overlap in some cases. Therefore, we made the networks for two small cube inputs shared with the same weight to reduce overfitting and designed the classification architecture as dual branch, merging global and local characteristics to enhance feature representation (Fig. 2). Features extracted from the vesicoureteral junction cube and the peri-stone ureter inputs were summed and concatenated with the features extracted from resized entire ureter input (Fig. 2).

The efficacy of the modified 3D version of convolutional neural network (CNN) (ResNet3D [17] and DenseNet3D [18]), ResNet3D) with mixed 3D–2D convolution [19] and 3D transformer (TimeSformer [20]) were compared. The Data Supplement contains more information on the neural network architecture of each model and training details. A class activation map was created to identify the discriminative structures, which influenced the deep-learning model to make the predictions. Our analyses were based on Python 3.6 (Python Software Foundation, Wilmington, DE) and the pytorch1.8 framework was applied to develop the models.

Ureteral wall thickness and HU density of the ureter under the stone measurement

The CT images were visualized using ITK-SNAP 3.8. Two experienced radiologists measured the ureteral wall thickness and HU density of the ureter under the stone according to previously described methods [9, 14]. Both radiologists were aware of the clinical history of the ureteral stone, but were completely blinded to the accessibility. If radiologists cannot distinguish ureteral thickness, it was considered to have a negative test result. If the difference between the two radiologists was > 30%, the two radiologists read together to reach a consensus. Otherwise, the mean value was used.

Statistical analysis

Demographic characteristics, stone characteristics, and surgical data were displayed as frequencies and means as appropriate. Th *T* test was used to compare the continuous variables and Fisher exact test to analyze differences in categorical variables between the control and narrow ureter groups. The classification performance of both the internal and external test sets was evaluated using the classification accuracy, precision, recall (sensitivity), specificity, F-measure (F1 score), and AUC of ROC score. The cutoff value for ureteral wall thickness and HU density of the ureter under the stone was chosen according to the maximum Youden index. The cutoff values for the external test set were recalibrated to balance specificity, sensitivity, and precision. The Delong test was used to compare different ROC AUCs. A P value < 0.05 was considered statistically significant. The clinical utility of the model is determined using decision curve analysis. Statistical analysis was performed using R version 3.2.3 software (http://www.r-project.org/).

Results

Disease and patient characteristics

A total of 1596 and 311 cases were included in the control and narrow ureter groups in the Shanghai General Hospital cohort, respectively, which were split into the train, validation, and test set. A total of 70 and 12 cases were included in the control and narrow ureter groups in the Jiangqiao Hospital of the Jiading district of Shanghai used for external independent validation. The demographic data (age and gender) and stone laterality were comparable between the control and narrow ureter groups in the two cohorts (Table 1). There is a low percentage (24/1907, 1.3%) of proximal stones in the Shanghai General Hospital cohort, while the percentage of proximal stones is higher in the external test set due to different surgical preferences. Most of the three input regions (peri-stone ureter, vesicoureteral junction, and entire ureter inputs) labeled by the program can be used directly, but we noticed that an empty bladder led to a relatively low accuracy of bladder segmentation by OBELISK-Net; thus, corresponding vesicoureteral junction inputs require adjustments (Supplemental Table 3). Some other samples also required manual adjustment due to the anatomic variations (Supplemental Table 3).

Integrating the images of different parts of the ureter as inputs to produce a better predicting model

It was found that simply training the network using regional CT images such as inputs including the ureter around the stone or where the ureter enters the bladders can produce a weak classifier for the narrow ureter. After simply screening common deep-learning models (data not shown), DenseNet3D was selected as the backbone. Using a vesicoureteral junction cube, with intramural ureter, vesicoureteral junction, and part of the lower part of the ureter as the input, the model gave AUC of the ROC curve of 0.686 and an F1 score of 70.77%. Using a peri-stone ureter cube which includes stone and surrounding ureter as the input, we can archive an AUC of the ROC curve of 0.766 and an F1 score of 70.97%. We found dual-branch architecture, which receives local ureteral features from the vesicoureteral junction, and peri-stone ureter inputs and global contextual information from entire ureter inputs, and achieved a much higher AUC (0.884) and accuracy (85.29%). (Table 2).

Table 1 Demographic data

Groups	Shanghai General Hospital cohort (the train, validation, and test set)		
	Control ($N = 1596$)	Narrow ($N = 311$)	P value
Age: mean (SD)	52.37 (14.02)	53.56 (13.88)	0.17
Gender, n (%)			0.84
Male	1087 (68.11)	210 (67.52)	
Female	509 (31.89)	101 (32.47)	
Lateral, n (%)			0.455
Left	889 (55.70)	166 (53.38)	
Right	707 (44.30)	145 (46.62)	
Stone size: mean (SD)	6.89 (2.96)	6.97 (3.50)	0.66
	Jiangqiao Hospital in Jiading district of Shanghai (the external test set)		
	Control ($N = 70$)	Narrow ($N = 12$)	P value
Age: mean (SD)	45.24 (14.50)	45.50 (18.13)	0.95
Gender, n (%)			0.28
Male	51 (71.42)	11 (93.3)	
Female	19 (28.57)	1 (6.67%)	
Lateral, n (%)			0.35
Left	37 (52.9)	4 (33.3)	
Right	33 (47.1)	8 (66.7)	
Stone size: mean (SD)	6.72 (2.42)	5.25 (2.01)	0.05
Proximal stone, n (%)	24 (34)	8 (66)	0.05

SD standard deviation

Comparison of different neural networks

The accuracies, precisions, recalls (sensitivities), specificities, and F1 scores for the prediction of narrow ureter using the deep-learning models are listed in Table 2. It was observed that the DenseNet3D-based model achieved an accuracy of 85.29% and AUC score 0.884, followed by the ResNet3D-based model (accuracy: 77.94%, AUC ROC: 0.824), ResNet mixed convolution (MC) (accuracy: 77.94%, AUC ROC: 0.819) and the TimeSformer-based model (accuracy: 76.47%, AUC ROC: 0.773) (Table 2) (Fig. 3). Furthermore, the external test set confirmed the performance with AUCs of 0.824 (95% CI 0.650–0.998) for the DenseNet3D-based model (Table 2). The cutoff values were adjusted to 0.72, 0.55, 0.58, and 0.21 for neural networks DenseNet3D, ResNet3D, ResNet3D MC, and TimeSformer, which give relatively balanced specificity, sensitivity, and acceptable precision. The confusion matrixes for DenseNet3d in the internal and external test sets are shown in Fig. 4. A subset analysis of proximal ureteral stone in the external test sets revealed an AUC of 0.782 (95% CI 0.597–0.967) for the DenseNet3D-based model.

Models with no proximal stone were also trained and tested in datasets without proximal stones (Table 3). The results were consistent with those of proximal stones, and the AUCs of most candidate predictors were slightly lower than those of models with proximal stones (Table 3). DenseNet3D-based model archived an AUC of 0.800 (95% CI 0.683–0.916) in the external test set, the only predictor whose lower 95% CI surpassed 0.5.

Neural network outperforms other candidate predictors

Although previous studies showed that a previous ureteroscopy or a history of double-J stent placement [2, 4] and older age are associated with a higher success rate of access in a population of mainly proximal and renal stones, the correlation between age and narrow ureter is not statistically significant in both the Shanghai General Hospital cohort and external test set (both $P > 0.05$). A previous ureteroscopy or a history of double-J stent placement was correlated with narrow ureter (OR 1.73 95%CI 1.16–2.58, $p = 0.007$) only in the Shanghai General Hospital cohort. However, it was not correlated with narrow ureter ($P > 0.05$) in the external test set due to a low incidence of a previous ureteroscopy or a history of double-J stent placement ($N = 1$). Thus, the predictor of a previous ureteroscopy or a history of double-J stent placement was only applied in the internal test set and got an AUC score of 0.54. The DenseNet3D-based model outperformed the AUC score of a previous ureteroscopy or a history of double-J stent placement ($P = 6.386 \times 10^{-10}$).

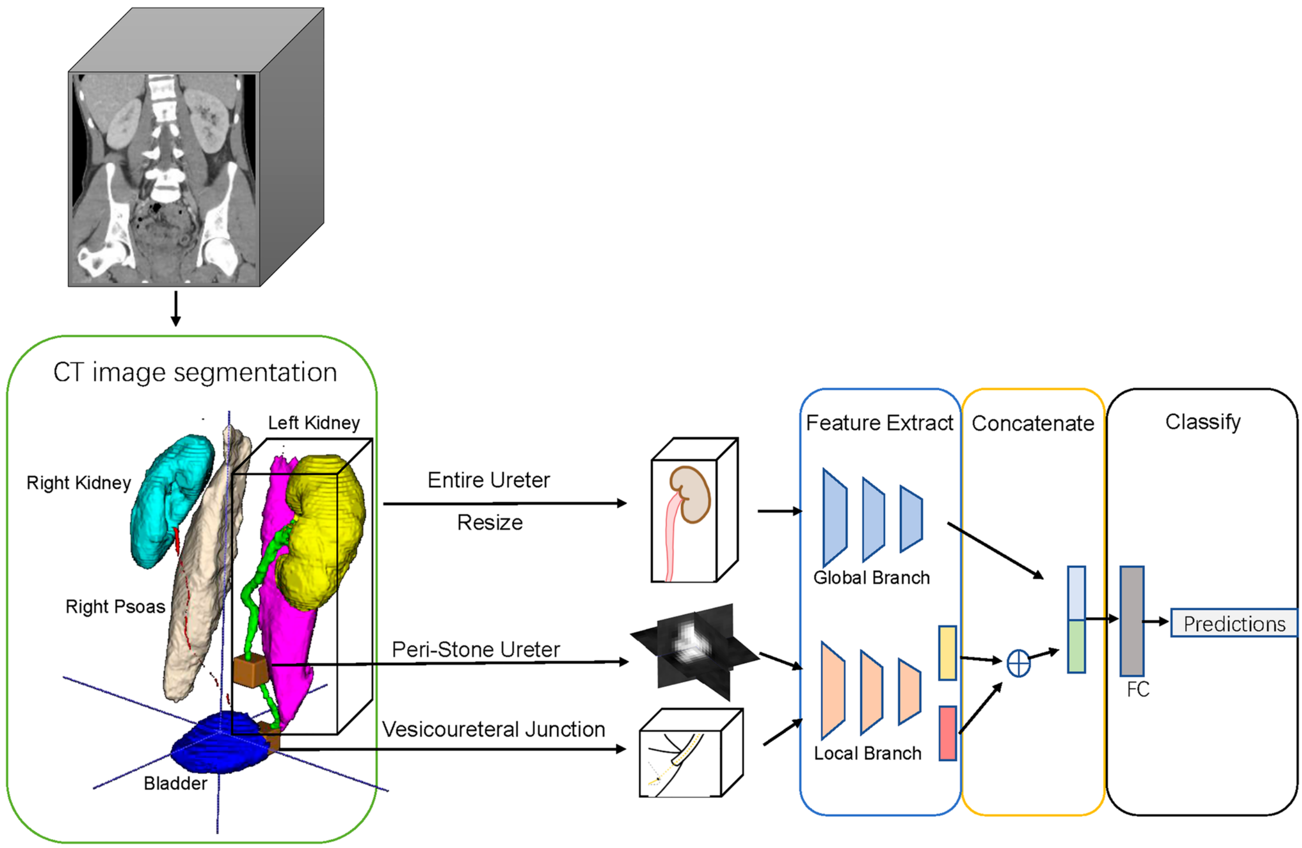

Fig. 2 The process of getting the volume of interest and representative architecture of the neural network. The morphological information of the bladder and psoas muscle was obtained from ensembled abdominal segmentation using OBELISK-Net and U-Net. The ureteral stone was segmented using the previously described method. Vesicoureteral junction, peri-stone ureter, and entire ureter inputs were obtained using the aforementioned anatomical information. The vesicoureteral junction and peri-stone ureter input features were summed and concatenated with the downsized entire ureter input features. Note: ureter in this representative figure is manually depicted, not from the segmentation algorithm. *FC* fully connected layers

The DenseNet3D-based model outperformed the AUC score of ureteral wall thickness (0.623 and 0.573) and HU density of the ureter under the stone (0.641 and 0.554) in the internal and external test sets (Table 2; Fig. 3a–b). The differences between the DenseNet3D-based model, ureteral wall thickness, and HU density of the ureter under the stone were statistically significant, with $P=0.014$ and $P=0.003$ for the internal test set, respectively. The differences between the DenseNet3D-based model, ureteral wall thickness, and HU density of the ureter under the stone were also statistically significant with $P=0.046$ and $P=0.048$ for the external test set.

Clinical utility of the neural network model

In the internal and external test sets, the decision curve analysis revealed that using the DenseNet3D-based model to predict adds more net benefit than either the "treat all" or "treat none" strategies scheme for a wide range of threshold probability, indicating the clinical usefulness of the neural network model (Fig. 3c–d).

Establishing a neural network that predicts both narrow ureter and ureteral flexion

During the same period of Shanghai General Hospital, Jiangqiao Hospital in Jiading district of Shanghai cohorts, we identified nine and two other patients with the additional criteria whose access to proximal ureteral stones were difficult because of not a narrow ureter, but ureteral flexion and switched to a flexible ureteroscope treatment. Although the incidence of impassable ureteral flexion is low, we attempted to fine-tune the previously trained narrow ureter DenseNet3D-based network to predict both narrow ureter and ureteral flexion. Extra oversampling with data augmentation was applied to emphasize ureteral flexion. The internal test set was formed by only narrow ureter cases. The AUC of the DenseNet3D-based network is 0.872 (95%CI 0.785–0.961), which was similar to the results from the

Table 2 Prediction of a narrow ureter

Comparison purpose	Model	AUC (95%CI)	Accuracy (%)	F1 Score (%)	Specificity (%)	Recall (sensitivity) (%)	Precision (%)
Comparison of different architectures with DenseNet3D	Dual branch	0.884 (0.797–0.971)	85.29	84.38	91.18	79.41	90.00
	Single branch						
	1. Vesicoureteral junction input	0.686 (0.546–0.809)	72.05	70.77	76.47	67.65	74.19
	2. Peri-stone ureter input	0.766 (0.652–0.884)	73.52	70.97	82.35	64.71	78.57
	3. Entire ureter input	0.690 (0.561–0.819)	68.65	61.82	85.29	51.52	77.27
Comparison of different candidate predictors in the internal test set	DenseNet3D	0.884 (0.797–0.971)	85.29	84.38	91.18	79.41	90.00
	ResNet MC	0.819 (0.712–0.926)	77.94	77.61	79.41	76.47	78.79
	ResNet3D	0.824 (0.720–0.927)	77.94	75.41	88.24	67.65	85.19
	TimeSformer	0.773 (0.656–0.889)	76.47	73.33	88.24	64.71	84.62
	HU density of the ureter under the stone	0.641 (0.503–0.779)	64.71	63.64	67.65	61.76	65.62
	Ureter thickness	0.623 (0.497–0.749)	61.76	59.38	67.65	55.88	63.33
	History of ureteroscope surgery or stent placement	0.544 (0.482–0.606)	54.41	68.04	11.76	97.06	52.38
Comparison of different candidate predictors in the external test set	DenseNet3D	0.824 (0.650–0.998)	Not applicable†	68.97	90.00	83.33	58.82
	ResNet MC	0.796 (0.674–0.918)	Not applicable†	45.16	82.86	58.33	36.84
	ResNet3D	0.794 (0.649–0.938)	Not applicable†	55.17	87.14	66.67	47.06
	TimeSformer	0.653 (0.488–0.819)	Not applicable†	41.67	67.14	66.67	30.30
	HU density of the ureter under the stone	0.556 (0.357–0.754)	Not applicable†	28.57	74.29	41.67	21.74
	Ureter thickness	0.573 (0.402–0.744)	Not applicable†	28.57	82.86	33.33	25.00

AUC area under the curve, *CI* confidence interval, †accuracy is not applicable for imbalanced data

original version (Table 4). The model trained with both narrow ureter and ureteral flexion correctly classified cases of ureteral flexion ($N=2$) in the external test set and archived an overall AUC of 0.837 (0.706–0.969) (Table 4). Thus, the prediction of both narrow ureter and ureteral flexion seemed feasible and did not compromise the prediction of the narrow ureter.

Network interpretation

Following that, we attempted to determine which portions of the image and which kind of characteristics contributed to the network's output. Class activation maps showed us the essential parts of the input picture that the neural network considered while making the final prediction. The representative class activation maps from the DenseNet3D-based model are shown in Fig. 5. The image features at the initial layers mainly reflect the structural information, such as the boundary of the ureter and stone, shape, and texture. On the other hand, features in the middle and deeper layers mainly represent high-level semantic ureter characteristics from anatomical images. These feature maps from the middle and deeper layers started to show activation on the intramural ureter and the ureter, where the bladder enters from images of the vesicoureteral junction. Furthermore, they focus on not only the ureter, but also the soft tissue surrounding the ureter.

Discussion

Deep learning is now a cutting-edge technology for image classification and segmentation because of its robust and accurate classification performance [21, 22]. The application of deep learning for medical use has been reported in prostate, bladder, lung, breast, colon, and eye diseases [21–26]. However, its application in the field of urinary calculi is still underutilized. We established and trained a network capable of accurately predicting narrow ureter from CT images. Firstly, the prediction helps doctors prepare ahead of time. Secondly, the option of active dilatation (coaxial dilator or balloon dilator) or passive dilatation (J-stent placement) and

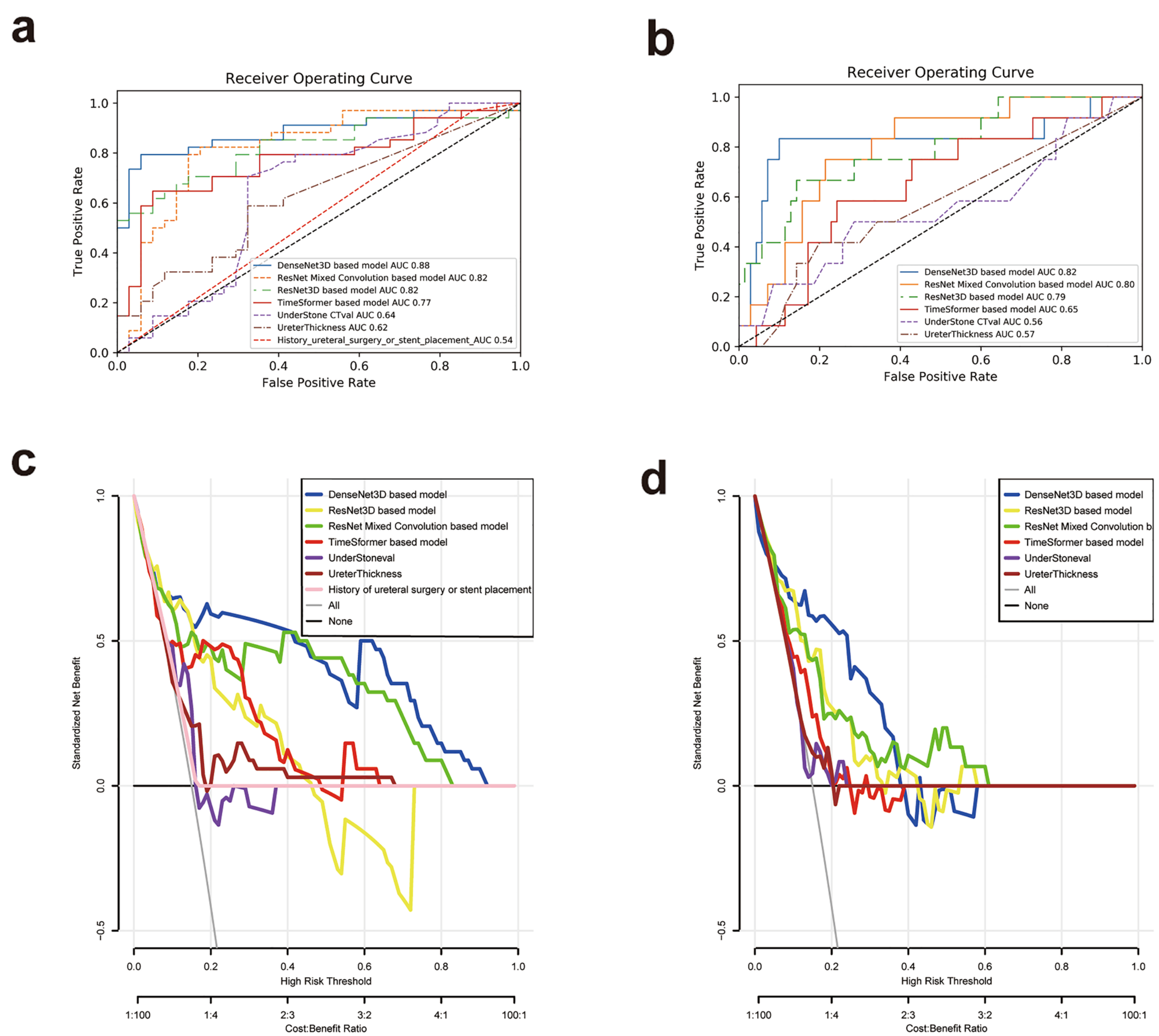

Fig. 3 Receiver operating characteristic (ROC) curves for candidate predictors. **a** Comparison of ROC curves for DenseNet3D, ResNet3D, ResNet mixed convolution, and TimeSformer-based models, under stone CT value, ureter wall thickness, and history of ureteral surgery or stent placement in the internal test set. **b** Comparison of ROC curves for DenseNet3D, ResNet3D, ResNet mixed convolution, and TimeSformer-based models, under stone CT value and ureter wall thickness in the external test set. **c** Comparison of decision curves for DenseNet3D, ResNet3D, ResNet mixed convolution, TimeSformer-based models, under stone CT value, ureter wall thickness and history of ureteral surgery or stent placement in the internal test set. **d** Comparison of decision curves for DenseNet3D, ResNet3D, ResNet mixed convolution, and TimeSformer-based models, under stone CT value and ureter wall thickness in the external test set. Understone CTval is short for the HU density of the ureter under the stone

potential risk can be informed. Informing patients of a specific surgical plan and scheduling in advance improve doctor–patient communication. Furthermore, advanced knowledge of the likelihood of narrow ureter can avoid excessive attempts to access the ureter and reduce the risk of various iatrogenic ureteral injuries and long-term complications such as ureteral stricture [27, 28].

Some retrograde access may be difficult for both rigid and flexible ureteroscopy. Previous studies are mainly based on populations with proximal ureteral and renal stones [1, 2]. There remains a paucity of data regarding the predictors of narrow ureters during semi-rigid ureteroscopic access. We found that the neural networks are better predictors than a previous ureteroscopy or a history of double-J stent placement and age.

Furthermore, we validated the models using external data. The DenseNet3D-based model generally performed well in CT scans from different data sources and exhibited

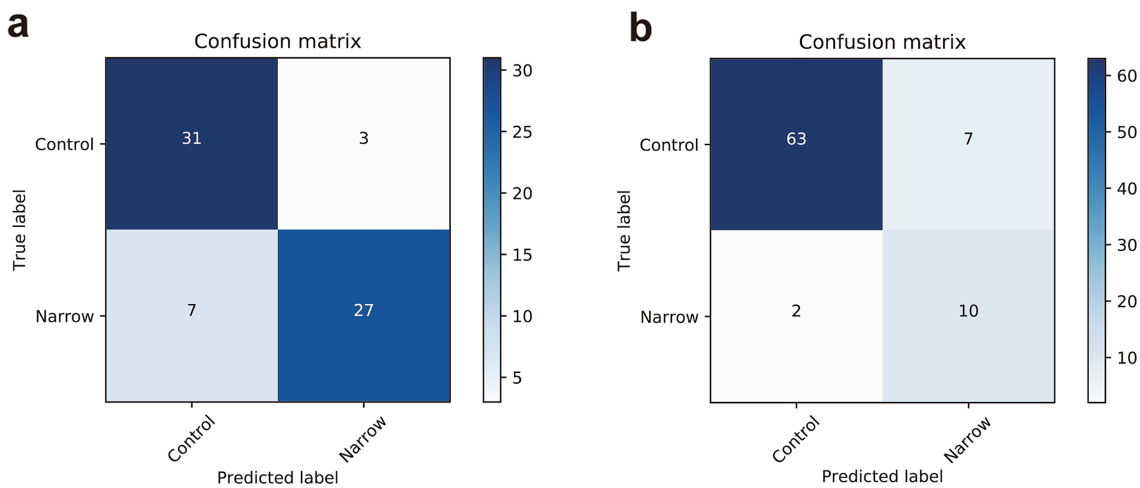

Fig. 4 Confusion matrix of the DenseNet3D model. **a** Confusion matrix of the DenseNet3D model based on the internal test set; **b** confusion matrix of the DenseNet3D model based on the external test set

Table 3 Comparison of different models trained without proximal ureteral stone

Comparison purpose	Model	AUC (95%CI)	Accuracy (%)	F1 Score (%)	Specificity (%)	Recall (sensitivity) (%)	Precision (%)
Comparison of predictors in internal test set	DenseNet3D	0.802 (0.688–0.915)	80.88	79.37	88.24	73.53	86.21
	ResNet MC	0.783(0.663–0.902)	77.94	77.70	77.94	77.94	73.17
	ResNet3D	0.760 (0.643–0.878)	72.06	67.80	85.29	58.82	80.00
	TimeSformer	0.732 (0.606–0.858)	72.06	67.80	85.29	58.82	80.00
	HU density of the ureter under the stone	0.641 (0.503–0.779)	64.71	63.64	67.65	61.76	65.62
	Ureter thickness	0.623 (0.497–0.749)	61.76	59.38	67.65	55.88	63.33
	History of ureteroscope surgery or stent placement	0.544 (0.482–0.606)	54.41	68.04	11.76	97.06	52.38
Comparison of predictors in external test set (without proximal ureteral stone)	DenseNet3D	0.800 (0.683–0.916)	Not applicable†	45.45	75.56	83.33	31.25
	ResNet MC	0.677 (0.323–1.000)	Not applicable†	57.14	91.11	66.67	50.00
	ResNet3D	0.718 (0.380–1.000)	Not applicable†	53.33	88.89	66.67	44.44
	TimeSformer	0.577 (0.323–0.832)	Not applicable†	28.57	73.33	50.00	20.00
	HU density of the ureter under the stone	0.705 (0.438–0.972)	Not applicable†	35.29	82.22	50.00	27.27
	Ureter thickness	0.561 (0.365–0.757)	Not applicable†	10.00	71.11	16.67	7.14

AUC: area under the curve, CI: confidence interval, †Accuracy is not applicable for imbalanced data

Table 4 DenseNet3D-based model that predicts both narrow ureter and ureteral flexion

Test set	AUC (95%CI)	Accuracy (%)	F1 score (%)	Specificity (%)	Recall (sensitivity) (%)	Precision (%)
Internal test set	0.872 (0.785–0.961)	80.88	79.37	88.24	73.53	86.21
External test set (with ureteral flexion)	0.837 (0.706–0.969)	Not applicable†	60.00	80.00	85.71	46.15

AUC area under the curve, *CI* confidence interval, †accuracy is not applicable for imbalanced data

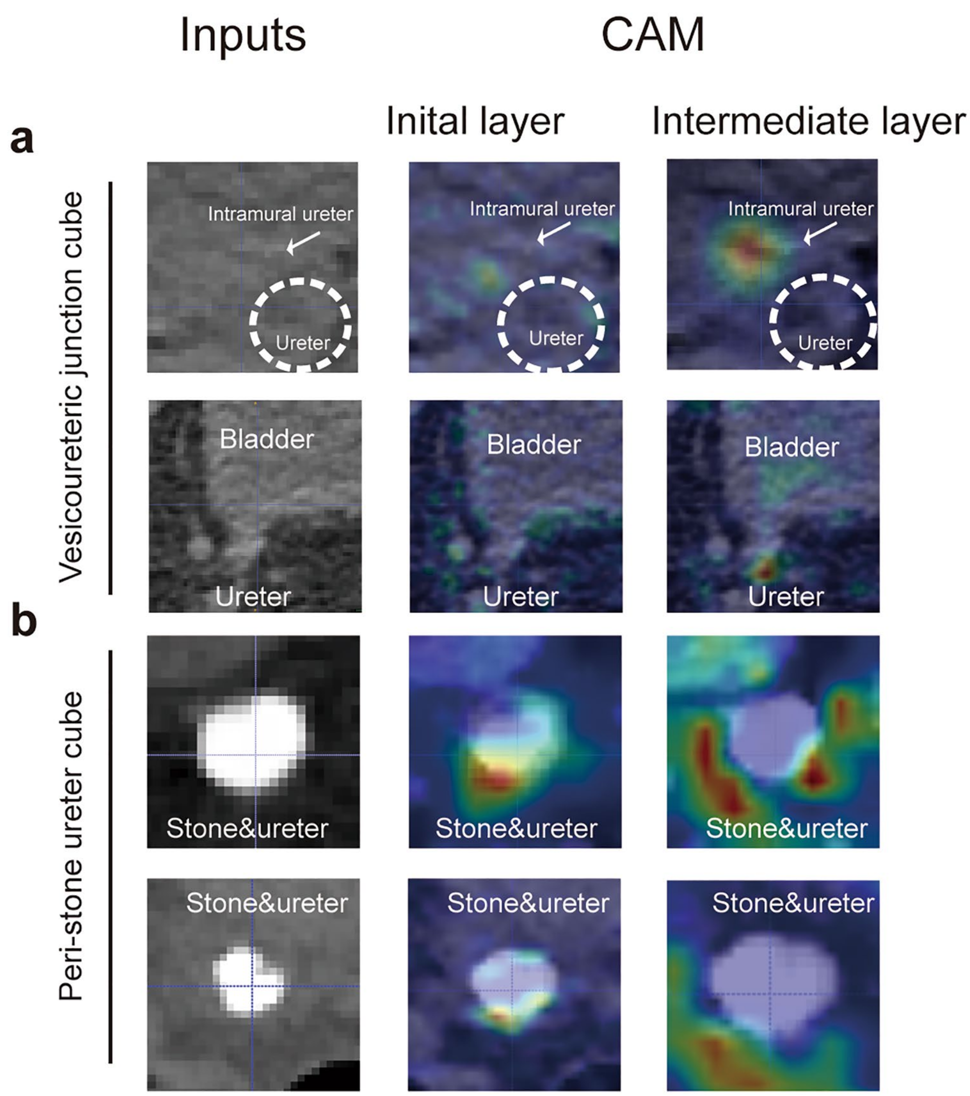

Fig. 5 Class activation maps for visualization. Input image (left) and CAM markup (right) for the narrow group from the vesicoureteral junction cube (**a**) and peri-stone ureter (**b**). CAM = class activation map

sensitivity (recall) of 83.33%, specificity of 90.00%, and slightly compromised precision. It indicated that the prediction is reliable if the classifier predicts a negative result. If the classifier predicts a positive result, there might be a small number of cases that are actually not as narrow as predicted. This trade-off fits real-world scenarios to reduce missed diagnoses. The decline of precision from the internal test set to the external test set is mainly because of the class imbalance of the external test set. Yet it did not change the fact artificial intelligence outperforms the HU density of the ureter under the stone and ureter wall thickness. Fine-tuning of the current model can be done when the model is applied to semi-rigid ureteroscopy of different diameters. Alternatively, the threshold can be adjusted by importance weighting. The prediction program can be deployed on most modern personal computers (16 GB memory and Linux operating system are required) without additional costs. Figure 3cd also indicates that its clinical application's benefits outweigh the cost.

One of the most challenging problems in computer vision is dealing with tiny objects. The ureters are only around 3–4 mm in diameter in a human adult. What is more difficult is that the feature difference between the two groups is subtle; even the radiologist could not distinguish them from the CT images. As ureters have different elasticity and angles, simply measuring the inner diameter of the ureter cannot simulate the real-world scene of the ureteroscope. If the whole set of scans is sent to a neural network, the classifier will be overwhelmed by background noise, and tiny ureter signals will be ignored. Direct segmentation of the ureter is not feasible, because some lower parts of the ureters are indistinguishable on non-contrast CT images due to limited resolution and similar HU values of soft tissue. Alternatively, we use anatomical knowledge to focus on two key areas, the lower part of the ureter and the ureter surrounding stone, and at the same time, pay attention to the global information of the entire ureter. This model mainly focused on narrow ureters during semi-rigid ureteroscope's access, because ureteral access sheaths have different diameters and rigidities compared with semi-rigid ureteroscopes, and a modified network structure (in progress) is preferred for flexible ureteroscope. Interestingly, modern transformer TimeSformer did not achieve comparable or superior performance compared with classical convolutional neural network DenseNet3D. It may be due to the training transformer requiring a considerable sample size, which does not often present in the real-world medical image sceneries.

Although the proportion of proximal ureteral stone in the training set is not high, since flexible ureteroscopy is now more recommended for proximal ureteral stone, our network model still had an acceptable prediction performance for overall proximal stone in the external test subset. Therefore, the model's generalization ability is well, and even for some primary medical institutions without flexible ureteroscopes, it is widely applicable. However, we should use this model with caution for ureteral stones above L3. Firstly, there were limited ureteral stones above L3 in the external test set to evaluate the predictive performance, as urologists preferred flexible ureteroscope for these patients. Secondly, even if the model does not predict a narrow ureter, the operation of a semi-rigid ureteroscope is still challenging due to its high position of proximal ureteral stones. It is sometimes difficult to access proximal ureteral stones by using a semi-rigid ureteroscope due to ureteral flexion. A switch to a flexible ureteroscope may be required. Using the deep-learning model to predict both narrow lumen and ureteral flexion may alleviate this situation. In addition, there may also be a narrow ureteral lumen between the peri-stone ureter cube and the ureter–bladder junctions. Our network may misclassify these patients. Theoretically, CTU can be used to explore the algorithm for the segmentation of the entire ureter, which may avoid this problem. However, clinically, patients with middle and distal ureteral stones are more often subjected to non-contrast CT rather than CTU.

In comparison with other candidate predictors, the neural network provides a more general solution. Firstly, although previous studies have shown that clinical parameters including a history of previous ureteroscopy or double-J stent, and older age are associated with a higher success rate of access [2, 4], these two predictors do not perform well in both internal and external test sets. Patients in our external test set are relatively young and generally lack active medical history, which makes a history of previous ureteroscopy or double-J stent ineffective. The internal test set comprised mainly distal/middle ureteral stones rather than proximal ureteral/renal stones, which was different from the population of the previous study. Secondly, the interpretability of ureteral thickness is better than that of the neural network, but a considerable part of ureteral thickness cannot be measured under CT (58.8% of the narrow group and 38% of the control group cannot distinguish ureteral thickness in the internal test set), which limits its capacity. In addition, in some patients, narrow uncompliant ureter had no association with inflammation and stimulation from stone and did not have increased ureteral wall thickness. Therefore, the thickness of the ureteral wall may not help distinguish such patients. Thirdly, the HU value of the ureter under the stone is related to the selected location. Even if the chosen area is close to, but still does not overlap, the stone area, the HU value increases significantly. The measurement error is not satisfactory. Furthermore, its utility in distinguishing intramural stones is disputed. To a certain extent, our neural network extracts characteristics around ureteral stones and the ureter–bladder junction at the same time to avoid the difficulties mentioned above.

However, this study has some limitations. Performances of this neural network were still limited by sample size, especially the prediction of ureteral flexion, and larger data sets can further improve the predictive precision of neural networks. Prediction for both narrow ureter and ureteral flexion may give urologists vague expectations. Integrating different predictive models' results may be more informative for clinical use. The data preprocessing is semi-automatic. Manual adjustment is required. Prediction of the exact location of the narrow lumen is unavailable. An additional limitation is a subjective impression of narrow ureter by the different surgeons.

Conclusions

The network provides an individualized preoperative prediction of narrow ureter based on non-contrast CT images, which could be employed as part of a surgical decision-making support system for patients with ureteral stones.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00240-022-01341-2.

Acknowledgements We gratefully acknowledge Jun Junior Wang from Computer School, Beijing Information Science and Technology University for technical support.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by Jun Wang, Dawei Wang, Yong Wang, and Shoutong Wang. The first draft of the manuscript was written by Jun Wang and Yi Shao and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding This work was financially supported by grants from the Science and Technology Commission of Songjiang District (Grant

No.18sjkjgg13), Shanghai Pujiang Program (Grant No. 2020PJD046), Scientific and Technological Innovative Action Plan from Science and Technology Commission of Shanghai Municipality (20Y11904600).

Declarations

Competing interests The authors declare no competing interests.

Conflict of interest The authors declared that they have no conflict of interest.

Ethical approval Ethics approval was obtained from the Institutional Research Ethics Board of Shanghai General Hospital. All procedures performed in studies were in accordance with the institutional ethical standards and with the 1964 Helsinki Declaration.

Informed consent Informed consent was waived.

References

1. Kijvikai K, Haleblian GE, Preminger GM et al (2007) Shock wave lithotripsy or ureteroscopy for the management of proximal ureteral calculi: an old discussion revisited. J Urol 178:1157–1163. https://doi.org/10.1016/j.juro.2007.05.132
2. Viers BR, Viers LD, Hull NC et al (2015) The Difficult ureter: clinical and radiographic characteristics associated with upper urinary tract access at the time of ureteroscopic stone treatment. Urology 86:878–884. https://doi.org/10.1016/j.urology.2015.08.007
3. Ambani SN, Faerber GJ, Roberts WW et al (2013) Ureteral stents for impassable ureteroscopy. J Endourol 27:549–553. https://doi.org/10.1089/end.2012.0414
4. Mogilevkin Y, Sofer M, Margel D et al (2014) Predicting an effective ureteral access sheath insertion: a bicenter prospective study. J Endourol 28:1414–1417. https://doi.org/10.1089/end.2014.0215
5. Jendeberg J, Thunberg P, Liden M (2021) Differentiation of distal ureteral stones and pelvic phleboliths using a convolutional neural network. Urolithiasis 49:41–49. https://doi.org/10.1007/s00240-020-01180-z
6. Kobayashi M, Ishioka J, Matsuoka Y et al (2021) Computer-aided diagnosis with a convolutional neural network algorithm for automated detection of urinary tract stones on plain X-ray. BMC Urol 21:102. https://doi.org/10.1186/s12894-021-00874-9
7. Cummings JM, Boullier JA, Izenberg SD et al (2000) Prediction of spontaneous ureteral calculous passage by an artificial neural network. J Urol 164:326–328. https://doi.org/10.1016/S0022-5347(05)67351-X
8. Mishra AK, Kumar S, Dorairajan LN et al (2020) Study of ureteral and renal morphometry on the outcome of ureterorenoscopic lithotripsy: The critical role of maximum ureteral wall thickness at the site of ureteral stone impaction. Urology annals 12:212–219. https://doi.org/10.4103/UA.UA_95_19
9. Bulbul E, Ilki FY, Gultekin MH et al (2021) Ureteral wall thickness is an independent parameter affecting the success of extracorporeal shock wave lithotripsy treatment in ureteral stones above the iliac crest. Int J Clin Pract 75:e14264. https://doi.org/10.1111/ijcp.14264
10. Kachroo N, Jain R, Maskal S et al (2020) Can CT-based stone impaction markers augment the predictive ability of spontaneous stone passage? J Endourol 35:429–435. https://doi.org/10.1089/end.2020.0645
11. Guler Y, Erbin A, Kafkasli A et al (2021) Factors affecting success in the treatment of proximal ureteral stones larger than 1 cm with extracorporeal shockwave lithotripsy in adult patients. Urolithiasis 49:51–56. https://doi.org/10.1007/s00240-020-01186-7
12. Yamashita S, Kohjimoto Y, Iguchi T et al (2020) Ureteral wall volume at ureteral stone site is a critical predictor for shock wave lithotripsy outcomes: comparison with ureteral wall thickness and area. Urolithiasis 48:361–368. https://doi.org/10.1007/s00240-019-01154-w
13. Yoshida T, Inoue T, Omura N et al (2017) Ureteral wall thickness as a preoperative indicator of impacted stones in patients with ureteral stones undergoing ureteroscopic lithotripsy. Urology 106:45–49. https://doi.org/10.1016/j.urology.2017.04.047
14. Tran TY, Bamberger JN, Blum KA et al (2019) Predicting the impacted ureteral stone with computed tomography. Urology 130:43–47. https://doi.org/10.1016/j.urology.2019.04.020
15. Heinrich MP, Oktay O, Bouteldja N (2019) OBELISK-Net: fewer layers to solve 3D multi-organ segmentation with sparse deformable convolutions. Medical Image Anal 54:1–9. https://doi.org/10.1016/j.media.2019.02.006
16. Rister B, Yi D, Shivakumar K et al (2020) CT-ORG, a new dataset for multiple organ segmentation in computed tomography. Scientific Data 7:381. https://doi.org/10.1038/s41597-020-00715-8
17. He K, Zhang X, Ren S et al (2016) Deep residual learning for image recognition. IEEE. https://doi.org/10.1109/CVPR.2016.90
18. Huang G, Liu Z, Laurens V et al (2016) Densely connected convolutional networks. IEEE Computer Society. https://doi.org/10.1109/CVPR.2017.243
19. Du T, Wang H, Torresani L, et al (2018) 'A closer look at spatiotemporal convolutions for action recognition' IEEE/CVF conference on computer vision and pattern recognition
20. Gedas Bertasius HW, Lorenzo Torresani (2021) Is space-time attention all you need for video understanding? (Paper presented at the proceedings of the international conference on machine learning (ICML)). https://doi.org/10.48550/arXiv.2102.05095
21. Fenstermaker M, Tomlins SA, Singh K et al (2020) Development and validation of a deep-learning model to assist with renal cell carcinoma histopathologic Interpretation. Urology 144:152–157. https://doi.org/10.1016/j.urology.2020.05.094
22. Suarez-Ibarrola R, Hein S, Reis G et al (2020) Current and future applications of machine and deep learning in urology: a review of the literature on urolithiasis, renal cell carcinoma, and bladder and prostate cancer. World J Urol 38:2329–2347. https://doi.org/10.1007/s00345-019-03000-5
23. Sunoqrot MRS, Selnæs KM, Sandsmark E, et al (2021) The reproducibility of deep learning-based segmentation of the prostate gland and zones on T2-weighted MR images. Diagnostics 11:1690. https://www.mdpi.com/2075-4418/11/9/1690
24. Herrmann P, Busana M, Cressoni M et al (2021) Using artificial intelligence for automatic segmentation of CT lung images in acute respiratory distress syndrome (Methods). Front Physiol. https://doi.org/10.3389/fphys.2021.676118
25. Jiang Y, Yao H, Tao S, et al (2021) Gated skip-connection network with adaptive upsampling for retinal vessel segmentation. Sensors 21:6177. https://www.mdpi.com/1424-8220/21/18/6177
26. Chen Y, Ruan D, Xiao J et al (2020) Fully automated multiorgan segmentation in abdominal magnetic resonance imaging with deep neural networks. Med Phys 47:4971–4982. https://doi.org/10.1002/mp.14429
27. De Coninck V, Keller EX, Somani B et al (2020) Complications of ureteroscopy: a complete overview. World J Urol 38:2147–2166. https://doi.org/10.1007/s00345-019-03012-1
28. Dong H, Peng Y, Li L et al (2018) Prevention strategies for ureteral stricture following ureteroscopic lithotripsy. Asian J Urol 5:94–100. https://doi.org/10.1016/j.ajur.2017.09.002